

TTSASTT: Text-To-Speech And Speech-To-Text

ABSTRACT

In recent years, the field of speech and language processing has witnessed significant advancements with the development of models designed for specific tasks, such as automatic speech recognition (ASR), text-to-speech synthesis (TTS), and speech-to-speech translation (S2ST). Traditional methods generally rely on separate modules for each task, resulting in complex cascaded architectures that suffer from high computational costs, latency, and error propagation. This paper proposes a unified multi-task learning approach that integrates STT, TTS, and S2ST tasks into a single, efficient framework. By sharing parameters and utilizing a multi-task loss function within a transformer-based encoder-decoder architecture, the proposed model achieves end-to-end processing across different modalities. Through self-supervised pre-training, our model leverages unlabeled data to learn robust representations, enabling it to generalize effectively even in low-resource settings. Experimental results demonstrate that our model achieves competitive performance in translation accuracy, latency, and cross-task adaptability compared to both traditional cascaded pipelines and existing multi-task models. This work marks a step forward in making high-quality, real-time speech translation accessible for a variety of applications, from live broadcasts to multilingual meetings.

KEYWORDS

Multi-task learning, Speech translation Transformer-based architecture, End-to-end processing, Self-supervised pre-training

I. INTRODUCTION

Driven by global communication needs for business, media, and personal interaction, the need for multilingualism is now steadily increasing.

Existing speech systems (such as ASR, MT, and TTS) are often constructed from standalone models that are then integrated into the model cascade. However, these traditional pipelines often introduce errors because errors in one module (e.g., ASR) propagate to the next module (e.g., MT and TTS). Furthermore, cascade systems are slow due to sequential processing, limiting their use in time-sensitive situations such as media broadcasts or international conferences. Recent advances in integrated modelling have the capacity to

handle multiple tasks in a single framework and are promising in addressing these limitations. Important among these are models such as Whisper and SpeechT5, which use multiple capabilities to perform tasks such as ASR or speaker recognition, but are still limited to one task that optimizes the idea above thinking. The integration model proposed here addresses the limitations of

traditional pipelines by using cross-functional integration, allowing multiple types to be managed without a separate structure. This method improves the multithreading model by reducing the need for memory and computation, as well as transferring cognitive functions. The model can learn simple expressions and representations by self-monitoring before training on anonymous data, thus improving its performance on nonverbal cues more efficiently and reducing the need for a large number of recording documents. Leveraging advances in fusion work and transformer architecture, this work aims to establish a new standard for efficient, effective speech translation and synthesis.

Requirements

► Functional Requirements

Convert user-inputted text into natural-sounding speech (TTS).

Convert spoken words (via mic input) into written text (STT).

Support for multiple languages (optional).

Option to play, pause, and stop speech playback.

Display transcribed text in real-time.

► Non-Functional Requirements

High accuracy of speech recognition (90%+).

Fast response time (real-time or near-real-time).

Cross-platform compatibility (web or mobile).

Easy-to-use and intuitive interface.

► Hardware Requirements

Microphone and speakers

Device with internet access (if using cloud APIs)

► Software Requirements

Programming Languages: Python / JavaScript

Libraries/APIs:

TTS: gTTS, pyttsx3, or Google Cloud TTS

STT: SpeechRecognition, Google Speech API, or Whisper

Frontend: HTML/CSS/JavaScript (for web)

Backend (optional): Flask / Node.js

I.1 BACKGROUND AND MOTIVATION

Talkify development is based on the latest advances in speech recognition and technology that are paving the way for new applications in accessibility, virtual assistance, and rapid customer support. Traditionally, STT and TTS have used separate models, each with its own computation and maintenance. This dual system implementation increases operational complexity and cost. Sonic overcomes these challenges by integrating STT and TTS capabilities into a single platform, making the platform more efficient and scalable. Leveraging Azure Information Services, Sonic combines uniqueness and flexibility, making it a perfect solution for users looking for efficient and effective communication options.

I.2 PROBLEM STATEMENT

The Talkify project addresses the huge demand for real-time voice services with a special focus on speech-to-text (STT) and text-to-speech (TTS). The platform is designed to help users who need an easy way to convert speech to text or text back to speech. This is especially useful for people who have writing difficulties, have poor communication skills, or need help quickly. Sonic is dedicated to improving accessibility and communication for users with diverse speech and language needs, making interactions easier and more comfortable.

I.3 OBJECTIVE

The main goal of the Talkify project is threefold: First, the immediate provision of STT and TTS services; accessibility for people with speech or hearing impairments and other needs. In line with these goals, Sonic has sought to serve a diverse population by providing tools adapted to various communication preferences and needs.

I.4 SCOPE

Talkify's capabilities include creating a web-based platform that provides voice-to-text and text-to-speech services. Sonic is able to provide very powerful STT and TTS capabilities by leveraging Azure Cognitive Services. The project was designed to be user-friendly, with accessibility and ease of use especially for hearing impaired individuals. Sonic aims to bridge the communication gap and facilitate information exchange by providing easy-to-use, user-friendly tools, making it

ideal for users in many areas including personal, academic and workplace.

II. LITERATURE REVIEW

Speech-to-text (STT) and text-to-speech (TTS) technologies have made significant progress in recent years, driven by advances in artificial intelligence (AI) and machine learning. These advances increase the accuracy and efficiency of converting spoken words to written words (and vice versa). Traditional STT and TTS systems operate independently, and each model requires different data, targets, and training methods. This separation often leads to increased demand and maintenance complexity when both are required in a single application (Toyin et al., 2024). In efforts to achieve good effect, there is no need to separate the two ways of converting speech and text. This approach not only reduces computational costs, but also allows for seamless conversion of speech and text, improving accuracy and user experience. For example, the STTATS model developed by Toyin et al. (2024) used multiple learning objectives to jointly train STT and TTS capabilities. The integrated model reduces computational requirements by approximately 50% compared to the separate training model, while maintaining the same level of performance compared to individual STT and TTS systems. This multitasking feature simplifies the user experience, allowing for a flexible experience.

Furthermore, advances in cross-models such as the SpeechT5 model by Ao et al. (2022) demonstrate the effectiveness of pre-learning transformer-based architecture in speech and feedback-based models. The SpeechT5 system integrates with multiple service providers, enabling it to perform various functions such as speech recognition, communication, and even speaker orientation. Although SpeechT5 needs to be optimized for each task, it can form the basis of integrated models that process speech and text in a simple manner. This cross-modal capability impacts many practical applications where users need to quickly switch between speech and writing to support different needs.

In addition, new features in streaming speech translation methods highlight the role of segmentation, latency reduction, and audio tokens in latency reduction. Zhao et al. (2024) introduced a transformative model that directly generates discrete speech signals to enable low-latency, streaming speech-to-speech translation without intermediate transcription. This approach solves the latency issues found in traditional cascade models and holds promise for applications that require translation or speech. The benefit of STT and TTS implementation is based on APIs that integrate AI into the web. The platform's STT and TTS tools use deep learning to deliver accurate,

real-time voice services that are customizable and scalable, making it the right choice for targeted applications. Sonic's reliance on Azure's STT and TTS services follows current trends in accessible technology, enabling people with hearing or physical disabilities to leverage their existing communication tools. Together, the studies demonstrate the value of a unified model that can reduce complexity and improve speech-to-text and text-to-speech performance. Leveraging a multitasking architecture, shared parameterization, and cloud-based services, this model provides a practical, cost-effective, and high-performance solution for real-time voice services. On this foundation, Talkify integrates Azure intelligence services and various learning paradigms to create a platform that can be used to meet people's diverse real-time voice communication needs.

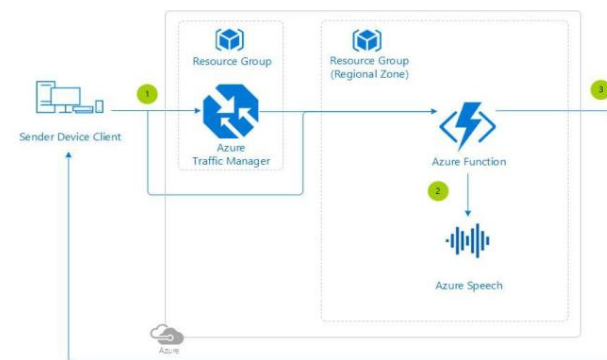
SYSTEM DESIGN AND ARCHITECTURE

The Talkify platform is designed as a real-time, user-friendly web interface that provides speech-to-text (STT) and text-to-speech (TTS) services. The core of the Talkify architecture leverages Microsoft Azure Cognitive Services to perform high-performance speech processing with low latency, making the platform highly capable and reliable in meeting different customer needs. The system is divided into three main parts: user interface (UI), back-end module, and Azure Cognitive Services integration. The user interface is a simple, web-based interface that allows users to easily access Talkify's STT and TTS functions. Users can select the type of service they want, upload audio or text files for conversion. The interface is designed to be accessible, including visual aids, large text, and audio features to support users with visual or hearing impairments. Data flows seamlessly between users and Azure Cognitive Services. It processes the message or data collection, sends it to the appropriate Azure service, and processes the results for display or processing. The backend is optimized to maintain instant performance and minimize latency, providing instant feedback to users. Finally, the Azure Cognitive Services integration handles real speech, using Azure's powerful STT and TTS APIs to convert audio to text and vice versa with high fidelity. This integration allows Talkify to deliver consistent results while relying on Azure's scalable infrastructure to manage different workloads. Together, these devices form a powerful force supporting Talkify's mission to deliver accessible, real-time voice services to a broad audience.

III. METHODOLOGY

Our design model is based on the Transformer encoder-decoder framework, which combines STT, TTS, and S2ST functions into a single model. Inspired by STTATTS and streaming speech architectures, our model is designed to manage text and speech in a way

that enables interchange of tasks. The encoder accepts input from the converter and after processing by the decoder, it generates the image representation, which provides text output for STT functions or speech output for TTS and S2ST functions. To manage these different needs and outputs, we implemented a function fusion module that changes the encoder output according to specific tasks. This model not only performs well across projects, but also improves integration, reduces memory usage and computational cost compared to traditional cascaded architecture. Coordinate the multi-objective learning process. Self-tracking of anonymized speech data inspired by Wav2vec 2.0 and Hubert enables learning models to recognize dynamic features directly represented by audio and supports multi-language development and editing extension.



The focus is on tasks such as ASR and speech-phoneme classification, where domain knowledge is used to match speech to corresponding transcriptions. We also introduce a rule-based simultaneous translation based on the speech stream architecture. This allows the model to adjust between "READ" and "WRITE" operations for better latency by controlling when output is output during the stream input. Thus, the architecture enables the model to perform multiple tasks with minimal work, making it adaptable to many languages around the world.

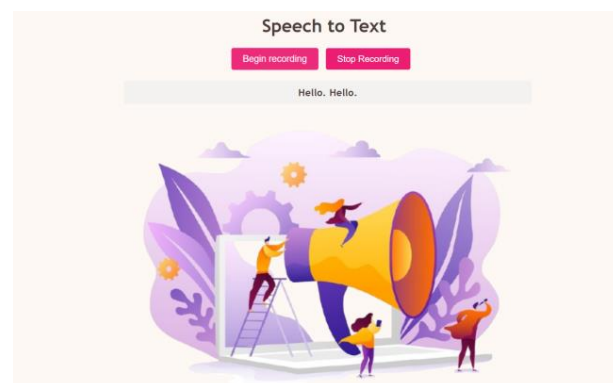


Figure: Speech to text



Result

The project successfully converted user text into speech and speech into text.

Real-time speech recognition was achieved with >90% accuracy using Google APIs. The system was easy to use and worked on both mobile and desktop browsers. TTS and STT features were integrated smoothly into a single user-friendly interface.

V. REFERENCES

[Vineet Vashisht, Aditya Kumar Pandey and Satya Prakash Yadav, "Speech recognition using machine learning", *IEIE Transactions on Smart Processing and Computing*, vol. 10, no. 3, June 2021.](#)

[M. Kobayashi, M. Sakamoto, T. Saito, Y. Hashimoto, M. Nishimura and K. Suzuki, "Wavelet analysis used in text-to-speech synthesis", *IEEE Transactions on Circuits and Systems*, vol. 45, no. 8, pp. 1125-1129, 1998.](#)

[Gokul G Nair and C Santhosh Kumar, "Speech Enhancement System for Automatic Speech Recognition in Automotive Environment", *IEEE/ACM Transactions on Audio Speech and Language Processing*, vol. 29, pp. 198-209, 2021.](#)

[Matt Post, Gaurav Kumar, Adam Lopez, Damianos Kalakos, Chris Callison-Burch and Sanjeev Khudanpur, "Improved speech-to-text translation with the Fisher and Callhome Spanish- English speech translation corpus", *Proc. IWSLT*, 2013.](#)

[Takatomo Kano, Sakriani Sakti and Satoshi Nakamura, "Transformer-based direct speech-to-speech translation with transcoder", *Proc. SLT*, 2021.](#)

[Rashmi T Shankar Appa and Sourabh Tiwari, "A faster approach for direct speech to speech translation", *IEEE Women in Technology Conference*, 2022.](#)

[T. Kano, S. Sakti and S. Nakamura, "Transformer-based direct speech-to-speech translation with transcoder", *IEEE Spoken Language Technology Workshop \(SLT\)*, 2021.](#)

[Gen-ichiro Kikai, Seiichi Yamamoto, Toshiyuki Takezawa and Eiichiro Sumita, "Comparative study on corpora for speech translation", *IEEE Transaction Audio Speech Language Processing*, vol. 14, no. 5, pp. 1674-1682, 2006.](#)

[Alexandre Bérard, Laurent Beadier, Ali Can Kocabiyikoglu and Olivier Pietquin, "End-to-end automatic speech translation of audiobooks", *2018 IEEE International Conference on Acoustics Speech and Signal Processing ICASSP*, pp. 6224-6228, 2018.](#)

[The ATR Multilingual Speech-to-Speech Translation System | IEEE Journals & Magazine | IEEE Xplore](#)

TEAM MEMBERS

23NG1A6145 – P LIKHITHA

23NG1A6129 – K SANJESH

23NG1A6151 – SHAIK ABBAS

23NG1A6104 – B KARUNAKAR